GOLD CLIP PACK

MAKE YOUR YOUR MASTERS SOUND LIKE GOLD



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01 | GOLD CLIP PACK: More Than a Clipper

GOLD CLIP PACK is more than just a clipper.

Gold Clip Pack is a family of plugins designed to bring loudness, saturation, and energy to your mixes and masters. It includes both a Mastering Clipper and a Track Clipper, powered by the same advanced internal technology. With unique features like the Gold loudness processor, Alchemy digital tape-style dynamics, Box Tone shaping, a True Parallel mixer, anti-overshoot oversampling, Clip Clock, Input and Output Trim Linking, and dynamic range visualization, Gold Clip Pack gives you the control to shape your mixes and masters with loudness and impact.

It's a simple tool, but underneath lies complexity. Yes, it's a clipper, but it's so much more than that. Once you fully understand Gold Clip, you'll unlock a wide palette of sounds—from aggressive and dirty to clean and pristine. Over the next few weeks, I'll send you a few email tutorials and videos to help you unlock the power of Gold Clip.

I promise that if you take a few minutes to read or watch each one, it will vastly improve the results you get from this unique tool.

What Is Gold Clip Pack?

Gold Clip is a clipper, loudness saturator, mid-to-high frequency peak processor, distortion device, box tone generator, clip clock, and true parallel mixer.

It can serve as a mastering processor, a unique dynamic effect for mixing, or a creative tool for production. Gold Clip can be clean and reserved or loud, aggressive, and distorted.

Gold Clip includes two loudness saturation algorithms, both called Gold, as well as a unique mid-to-high frequency peak processor called Alchemy. Both processors are integrated into the clipping algorithm to deliver a maximally smooth and loud master.

01 | GOLD CLIP PACK: More Than a Clipper

The Three Functions of Gold Clip

The Clipper is inspired by the overloading effects of two famous mastering converters. We named them Modern and Classic. There is also a clean clipper that we cooked up for you called Hard. These three clippers are like no other digital clippers out there. Try pushing your mixes into them and see how smooth and natural they sound. And the clipper isn't even the best part, but it is a necessary step to make Gold processing possible.

Gold processing uses a clever loudness technique that enhances quiet sounds in your mix, without altering the peaks. I like to call it compression without consequence, but in reality, it is compression without attack and release. Use the input trim to boost your mix up to the clipper's ceiling, add some Gold, and experience the power of this incomparable dynamic effect.

Alchemy is a peak processor that operates in the mid to high frequencies, reducing harshness and unwanted clipping effects. As the signal reaches the clip point, Alchemy gently softens the mid to high frequencies. I like to think of it as digital tape because it can take the ugly bite out of a mix when things get loud.

Technical Details

Gold Clip uses cutting-edge digital signal processing techniques to create a versatile, high-precision clipping effect with tons of flexibility and highly optimized performance. The primary clipping algorithms are modeled from a pair of well-known hardware units using integrated wave-shaping in conjunction with variable oversampling, resulting in a clipping sound that is clean, natural, and free of aliasing artifacts. Gold Clip also includes a pair of unique loudness saturation algorithms called Gold and a unique mid- and high-frequency peak shaper called Alchemy that work in conjunction with clipping algorithms to provide a maximally smooth and loud master. Additionally, all the signal processing algorithms achieve the best performance possible on modern CPUs by utilizing cache locality and data-level parallelism.

02 | UI Overview



1 - Box Tone High-Frequency Contour
2 - Modern, Classic & Hard Clipping
3 - Gold - Modern & Classic
4 - Clip Clock
5 - Oversampling Settings
6 - Input Short-Term LUFS
7 - Gold Processor
8 - Waveform Display
9 - Alchemy Processor
10 - Output Trim
11 - Delta
12 - Parallel Mixer
13 - Mixer Dry/Wet Link
14 - Tool Tips
15 - AB
16 - Input / Output Trim Link
17 - Input Trim
18 - Clipper Ceiling
19 - True Bypass

03 | THREE CLIPPERS

There are three settings for the clipper: Modern, Classic and Hard. These clippers can be described as soft, medium, and hard knee (respectively), but they are much more complex than that.

Modern - soft knee clipper
Classic - medium knee clipper
Hard - hard knee clipper
LPF - Integrated clipping low pass filter (LPF)

Modern and Classic are tuned to match the character and dynamic response of their hardware counterparts. Hard is a custom clipper that maintains the integrity of the signal up until the clip point. The Clip Filter (LPF) can be used when extreme amounts of clipping are necessary.

Clip Filter

The LPF button applies a slight low-pass filter to the signal to create a smooth, soft clipping sound. If linearity of the high frequencies is important, click this option off. If more aggressive clipping is required, turn this option on.

Clipper Off

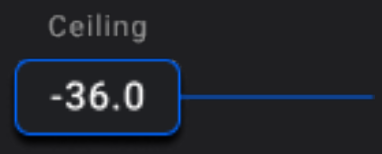
You can turn the clipper off by command clicking the clipper button. When the clipper is off, the ceiling needle turns grey and input meters turn red when the signal goes above 0.0dBFS. When the clipper is turned off, it is very important that the signal entering Gold Clip does not exceed 0.0dBFS. If the input signal exceeds 0.0dBFS, the Gold and Alchemy processors will cause distortions. Turning the clipper off allows you to use a limiter or clipper of your choice before Gold Clip. Or, use Gold Clip without a clipper or limiter at all and carefully adjust the input trim so the signal never exceeds 0.0dBFS.

Clipper in Practice

All three clippers sound quite different, and I encourage you to try different modes to see what works best for the song. Generally, I like to start with Classic and will switch to hard if I want the integrity of the signal maintained below the clip point. If I am doing extremely moderate clipping and using the gold and alchemy processing, I will choose the Hard setting. If I want low frequencies to bloom more, I will choose Modern.

Clip Ceiling

Sets the point at which the signal starts to clip. By default, it is set to 0.0dBFS. However, if you are working on track-based content at lower levels you can move the clip point down to the peak level of the source material. Both Gold and Alchemy processing work below the ceiling setting.



GOLD PROCESSING

Gold is fundamentally different than a compressor because it does not rely on attack and release times to perform compression. Gold performs a sample-by-sample analysis of the clipped signal and compresses the audio, leaving transients largely untouched, but increases sustain.



Gold compresses the audio but does so without the artifacts associated with attack and release on classical compressors. The compression is instantaneous, like the compression effect from the magnetization of tape.Gold processing relies on the input signal's level relative to the clip ceiling to perform compression. To access the full compression curve of Gold you must push the input signal as close to the clip ceiling as possible. However, the more aggressive of clipping you do, the less clean Gold processing will sound. It is suggested to push the signal as close to the clip point as possible but minimize clipping when using Gold for loudness. You can turn Gold processing off by command clicking the button or Gold amount knob.

Gold in Practice

For mastering I start with Modern Gold processing. This setting gives a maximum of 2.5 dB of gain and sounds clean and transparent. Gold Clip often sounds best when it is first in chain or just before your final limiter. Classic Gold Processing is much more aggressive sounding because it is applying more gain over a longer knee. Classic is better suited for signals with a wider dynamic range such as a drum bus, 808s or vocals, while modern is best suited for more dense content like a full mix.

04 | GOLD PROCESSING

Modern

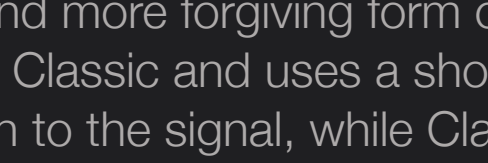
Modern is inspired by the soft-loudness saturation processing of the famous modern mastering converter. In effect, Modern is a gentler and more forgiving form of amplitude saturation. Modern applies less gain than Classic and uses a shorter knee. Modern can only apply a maximum of 2.5 dB of gain to the signal, while Classic can apply a total of 6 dB of gain (at the expense of more harmonic interaction). The maximum setting of the original iteration of the Modern hardware device displays +6 dB on the front panel, but the gain applied is only +2.5 dB.

Classic

Classic is inspired by the soft-knee loudness saturation of a popular mastering converter from the late 2000s. It uses a soft knee for its amplitude saturation but sounds more aggressive than Modern and causes more harmonic interaction at its maximum settings. The knee of Classic is fundamentally different from Modern because the knee starts lower when higher levels of saturation are applied.

Gold Unity Gain

Unity Gain only affects Gold processing. Unity Gain nulls the non-linear gain applied by Gold so that perceived loudness is constant, but peaks are reduced.

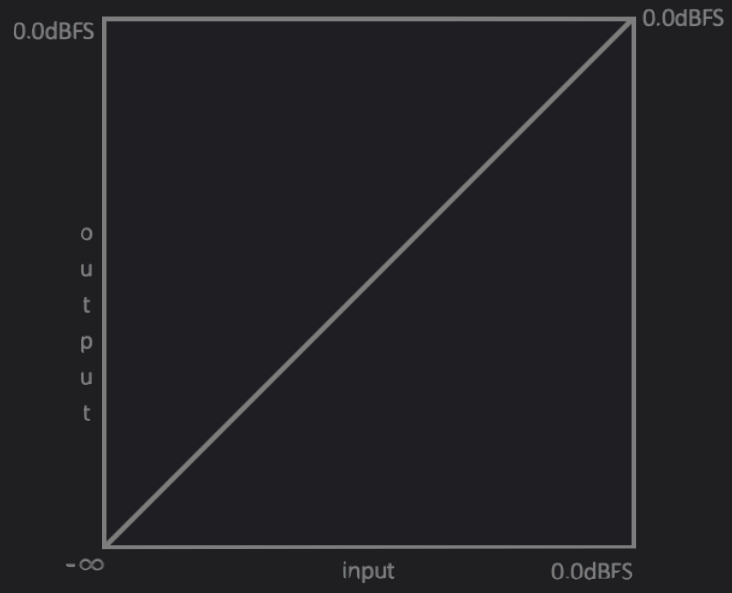


With Unity Gain turned on, the perceived loudness from Gold is maintained, but peaks are reduced. With Unity Gain turned off, the perceived loudness from Gold is increased, and peaks are left in place. For example, if Classic Gold is adding 6 dB of nonlinear gain, Unity Gain reduces the output of the signal by 6 dB as well. In effect, you are maintaining perceived loudness but lowering the peaks of the signal by 6 dB. Unity allows you to set the Gold type and amount without being misled by the changes in perceived loudness that Gold creates.

Unity Gain in Practice

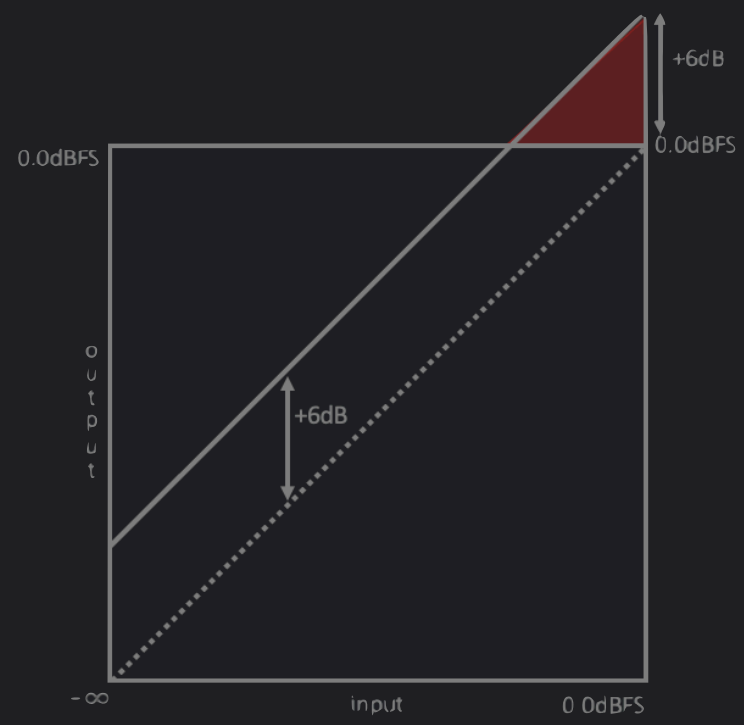
Turn Unity Gain on and set the amount and type of Gold processing that you want on your mix. By setting Gold with Unity Gain turned on you will listen to the dynamic effects of Gold and not be tricked by the loudness that it adds. After finding the best sounding Gold setting for the mix, turn Unity Gain off and enjoy the increased loudness that Gold provides.

How Gold Processing Works



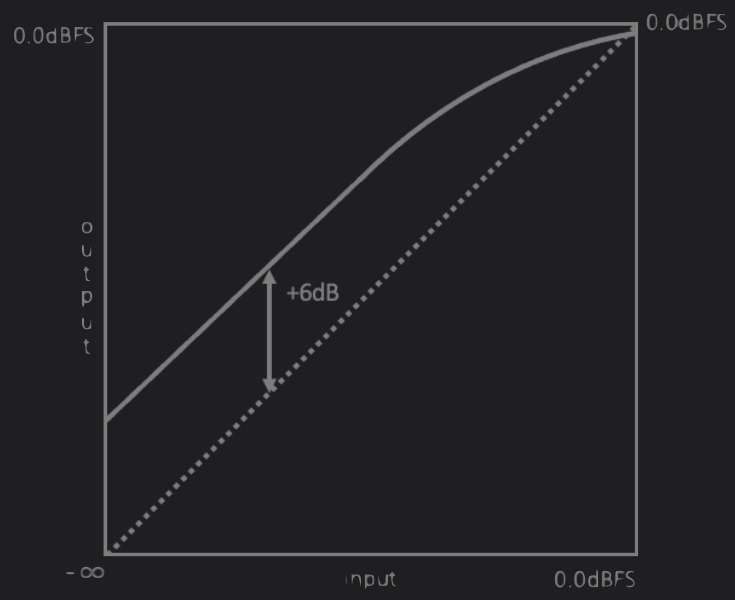
Transfer Function

If the input is equal to the output, it is considered a linear transfer. If the process changes as the input signal gets louder it is considered non-linear. The above graph shows a linear process — input on the X axis is equal to output on the Y axis.



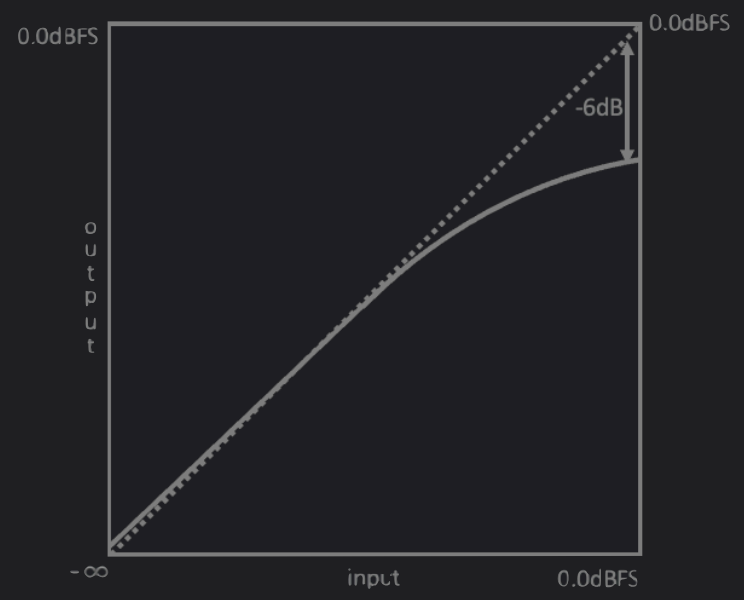
+6dB of Linear Gain

If you add 6dB of linear gain to the signal the output is 6dB louder and peaks are clipped by 6dB. Using a clipper in this way can yield creative results, but they are not always transparent.



Gold Processing

Gold processing adds up to 6dB of non-linear gain to the input signal. As the input signal approaches the clip ceiling, the non-linear gain will exponentially reduce to 0dB of gain. Gold processing makes the signal louder by amplifying low level material but keeps peak information below the clip point perfectly intact.

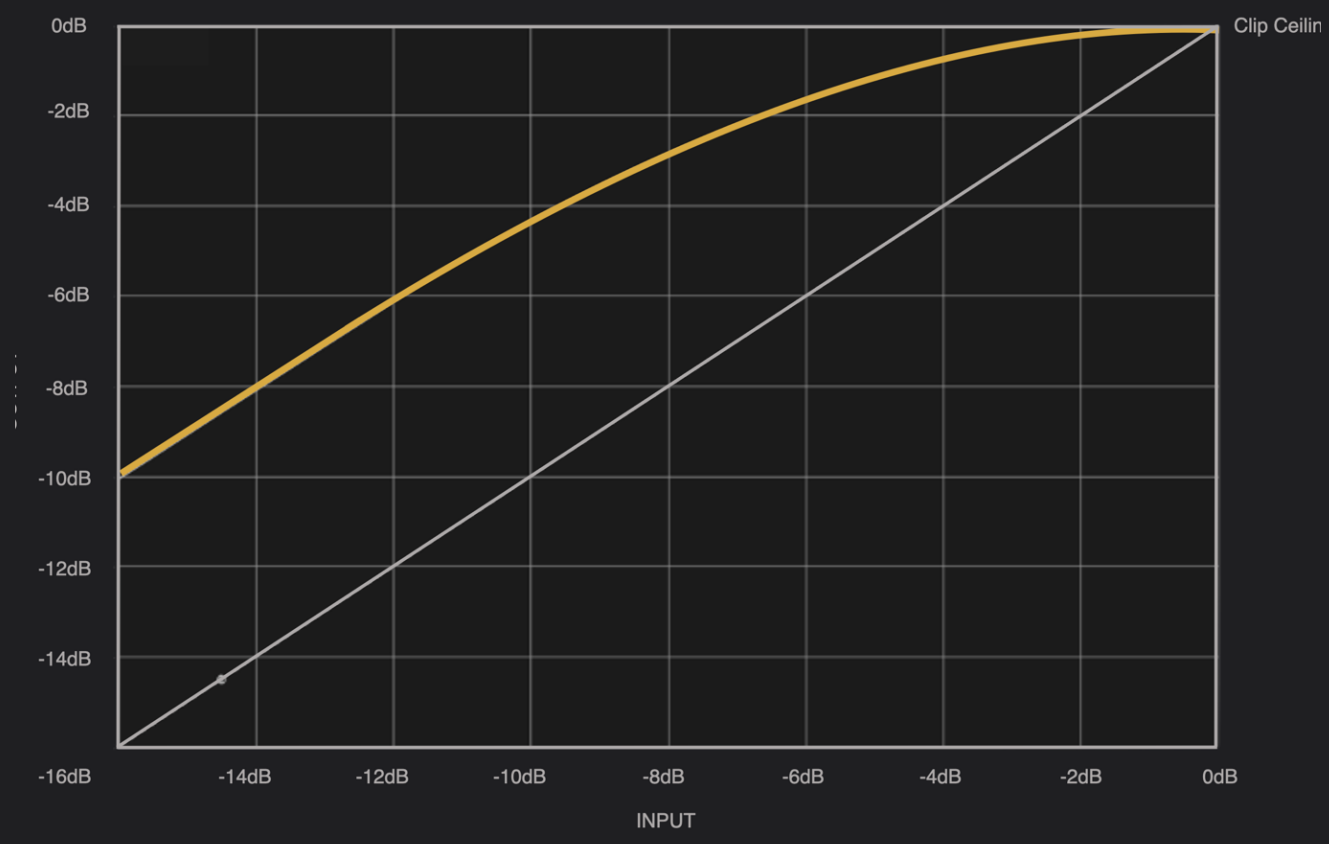


Unity Gain

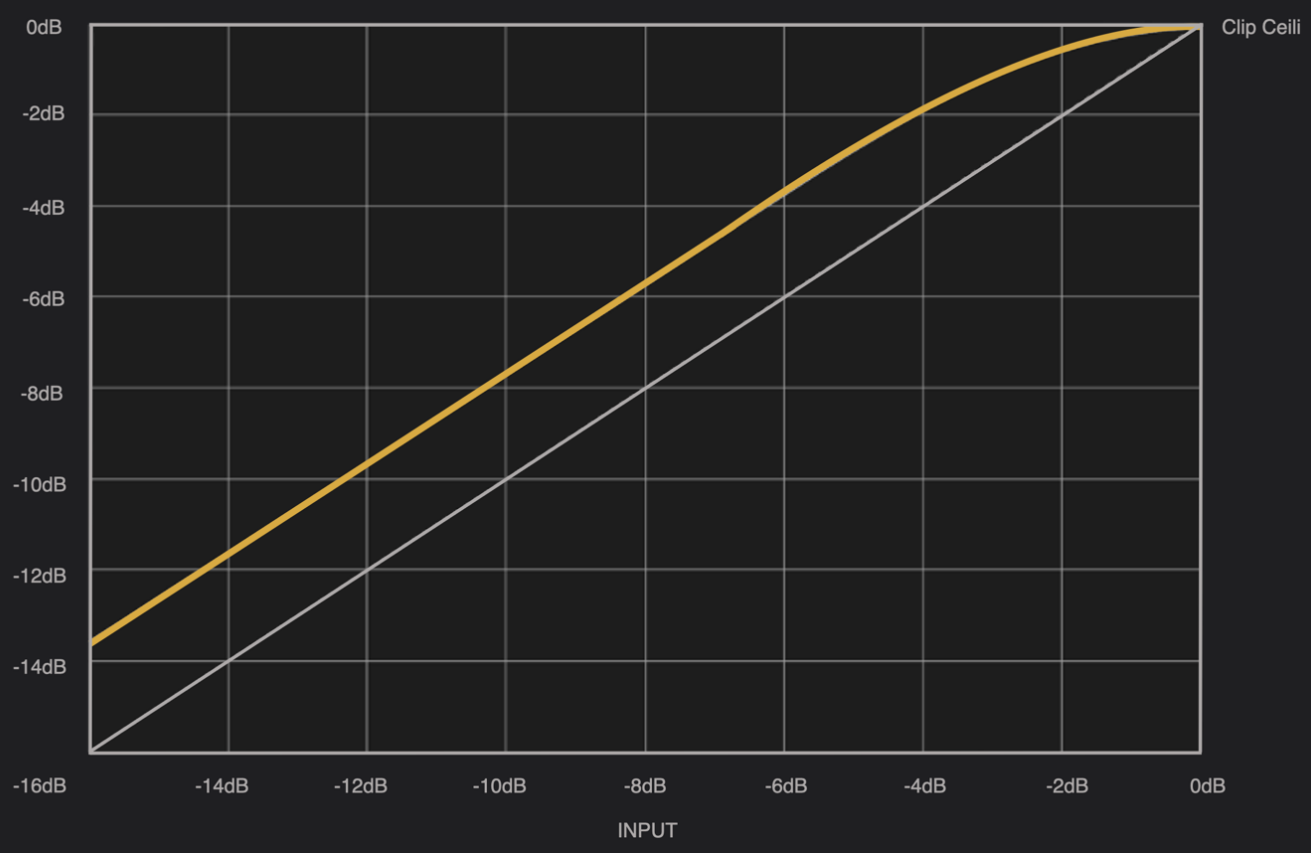
If Gold adds 6dB of level, Unity Gain will reduce the signal by 6dB after all other processing. This process reduces peaks by 6dB but maintains perceived loudness. Gold Processing with Unity Gain will increase your headroom without altering the integrity of your transients.

04 | GOLD PROCESSING

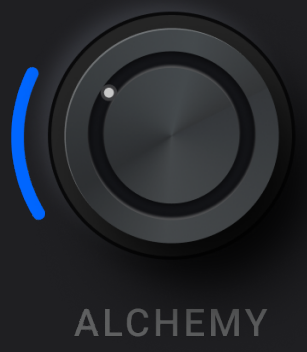
Classic Gold Processing (+6)



Modern Gold Processing (+2.5)



Alchemy 2 (GOLD CLIP ONLY)



When analog tape is overloaded, the high frequencies are reduced and softened, resulting in a more enjoyable experience during peak loudness. These loud bursts of energy become slightly darker, as the transient sharpness of the high frequencies is absorbed into the tape. Alchemy 2 emulates the characteristics of analog tape in a new and novel way. As sounds approach the clipping point, frequencies above 3 kHz are gradually reduced by up to 2dB. Unlike analog tape, Alchemy 2 preserves the digital detail of the low and low-mid frequencies while dynamically smoothing the upper midrange and top end at the loudest moments. Alchemy combines the top end smoothness of tape & the low frequency precision of digital in one processor.

Alchemy 2 in Practice

You can think of Alchemy as a subtle, gain-limited de'esser or high-frequency peak softener. First, set the amount of clipping you want and then adjust Alchemy to taste. If you struggle to hear Alchemy, try this: solo the snare drum and bring the clip ceiling down until you get about 1dB of clipping. Set Alchemy to 2dB and toggle it on and off by command-clicking the knob. You'll notice a subtle tone shift or warmth in each snare hit when Alchemy is active.

However, when a full mix is processed, the kick and bass are usually the loudest instruments. So, if a kick and crash cymbal hit the clip ceiling at the same time, Alchemy will quickly reduce the sizzle of the crash as the kick drum presses into the clip ceiling. Additionally, if you have a limiter adding more level after Gold Clip, that limiter will help give the perception of a louder kick because the volume relationship between the kick and cymbal changes during the kick drum. Again, it's subtle, but this unique frequency specific dynamic process is what makes Alchemy 2 sound like digital tape.

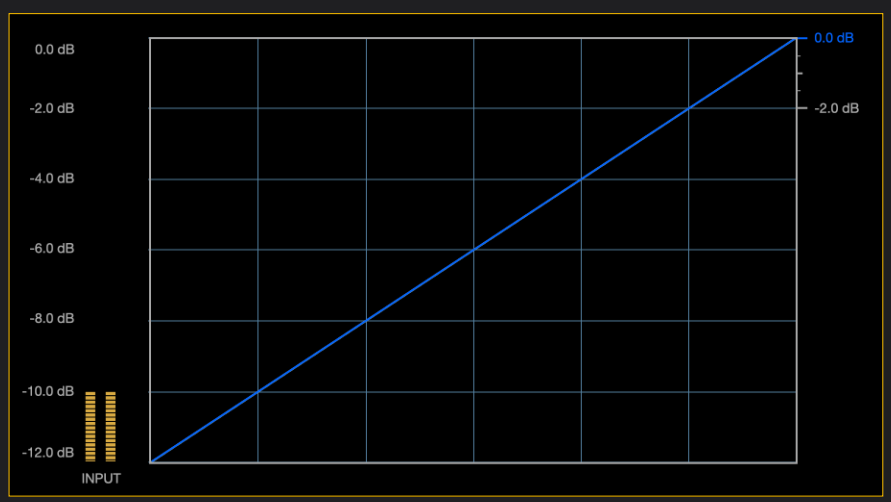
Turn up some Gold and Alchemy 2 on a mix and let me know what you think. I'm now using Alchemy 2 on more masters than ever.

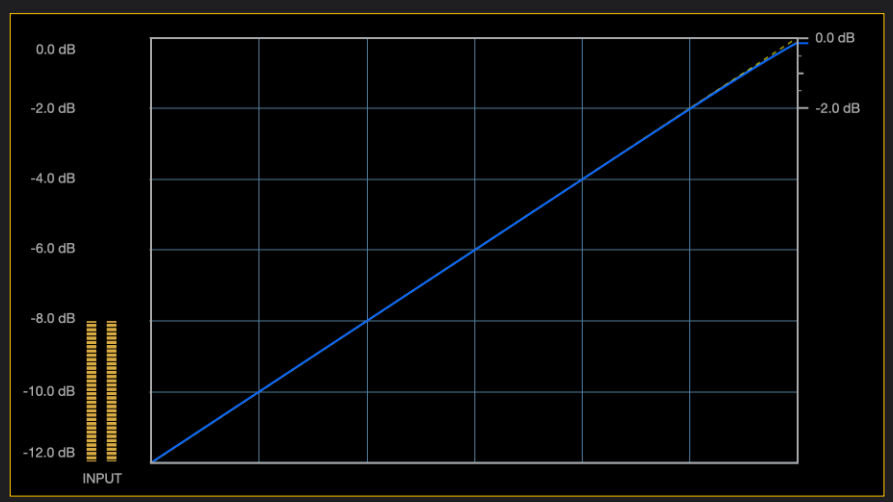
05 | ALCHEMY PROCESSING

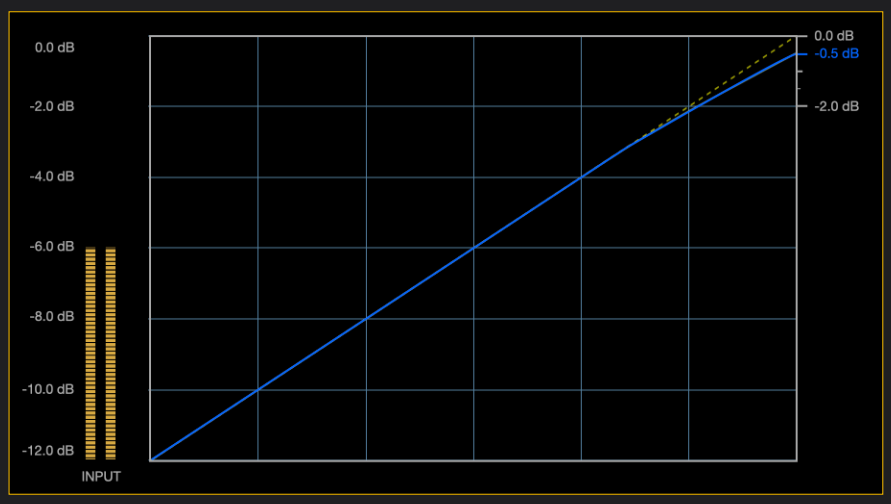
Alchemy reduces mid- and high frequencies by a maximum of 2dB

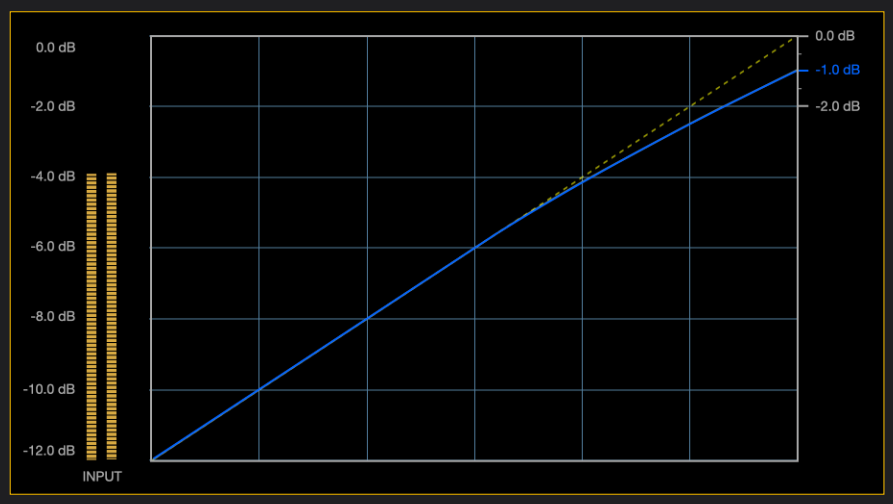


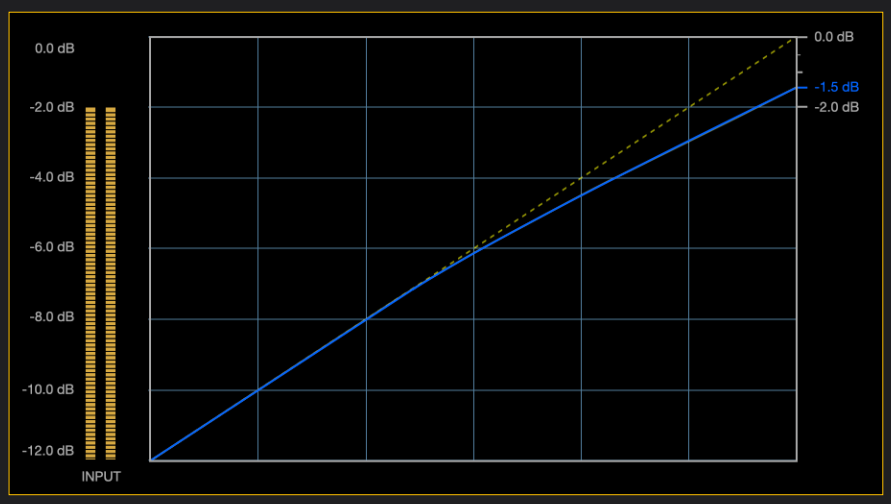
The below graphs show Alchemy set to 2.0dB at varying input levels with a 8kHz sine wav.

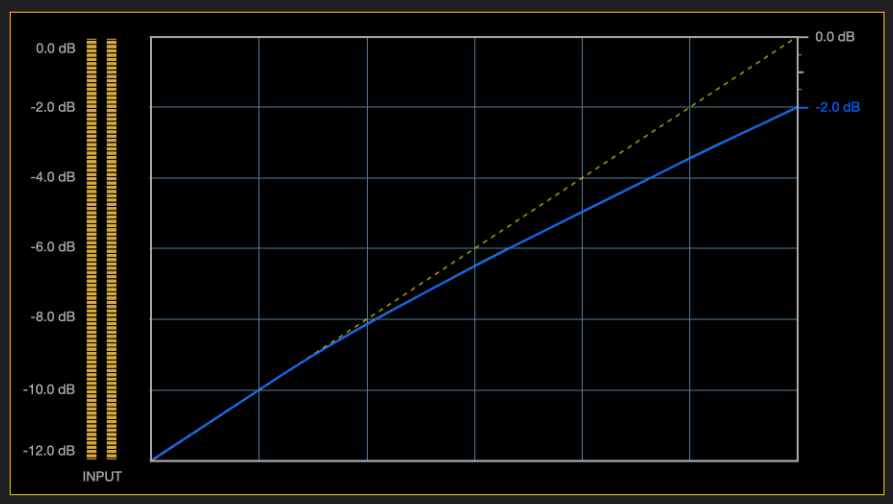












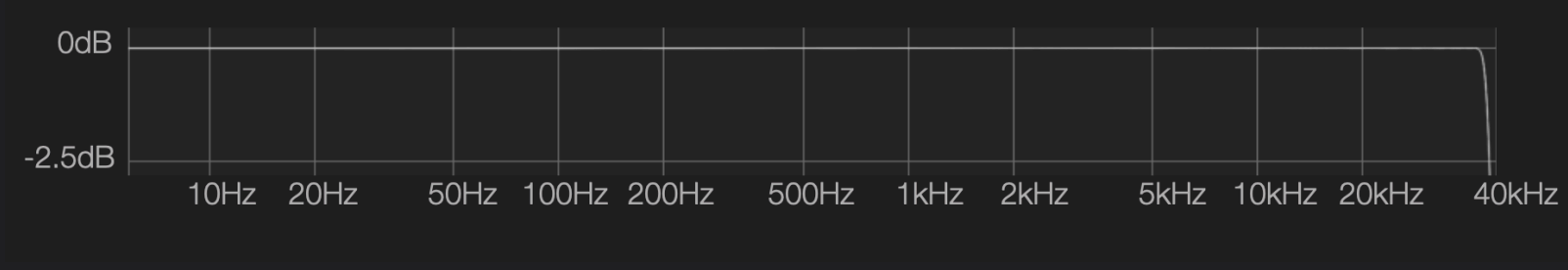
Box Tone (GOLD CLIP ONLY)

Box Tone is one of the more subtle processors in GOLD CLIP, but I often find it invaluable for smoothing out the top end of digital recordings. Think of it as a hyper-sonic low-pass filter that helps clean up some of the digital harshness in the very highest frequencies of a mix.

While subtle, Box Tone adds a slight contour to the highs, creating a smoother, more cohesive top end—especially when using the Modern setting.
Here's a look at what Box Tone is doing:

FLAT: The True Bypass Option

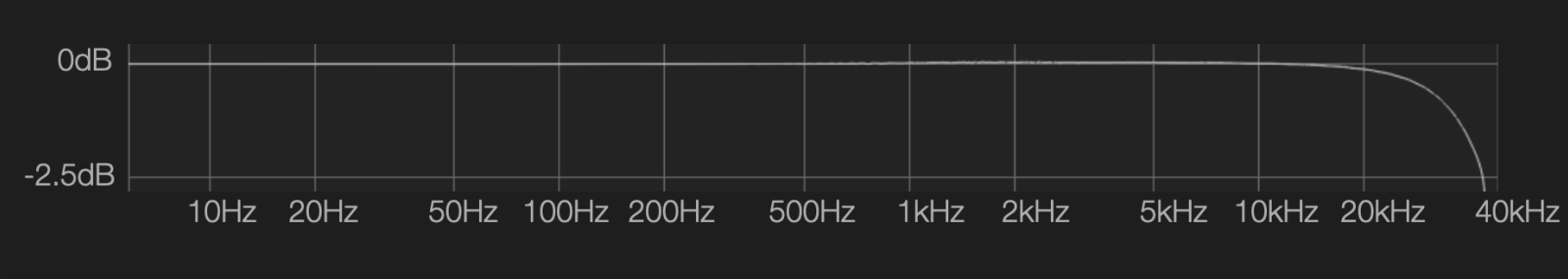
When set to FLAT, Box Tone is completely bypassed. There's no contour or hyper-sonic low-pass filtering applied—it remains true to the input signal all the way up to 40 kHz in a 96 kHz host sample rate session. This setting ensures a completely neutral pass-through, perfect for when you want an unaltered, transparent output.



CLASSIC: Smooth Air-Band Contour

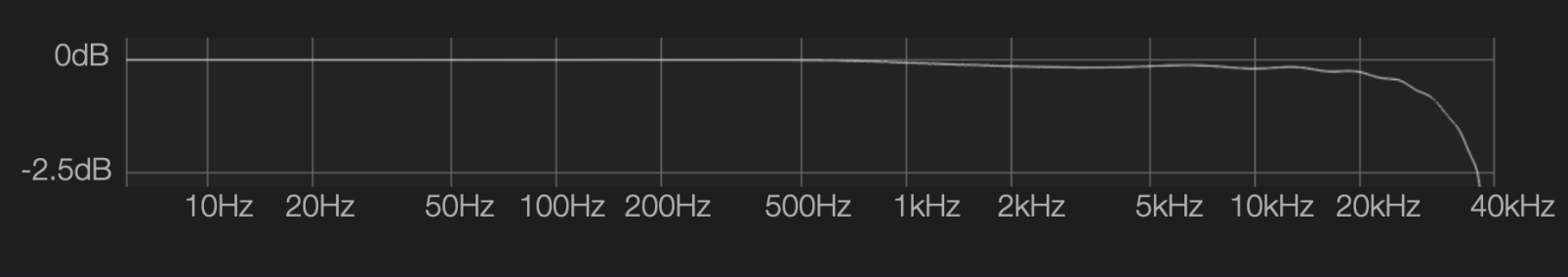
The CLASSIC setting adds a gentle air-band contour around 20 kHz, slightly softening the top end in your mix. It also applies a hyper-sonic low-pass filter, removing unwanted frequencies beyond our hearing range, between 20 kHz and 40 kHz.

This setting is perfect for subtly shaping the high end while keeping the extreme top end clean and true to input.



06 | BOX TONE PROCESSING

MODERN: High-End Shaping



The MODERN setting introduces a more aggressive contour than CLASSIC, with subtle cuts around 3 kHz and 8 kHz to reduce harshness and improve smoothness. It also slightly reduces the air-band around 20 kHz by 0.3 dB, adding a refined touch to the top end.

Like CLASSIC, it features a hyper-sonic low-pass filter to remove unwanted frequencies between 20 kHz and 40 kHz, but with a more assertive tonal shaping that's ideal for modern productions requiring a tighter, smoother high-frequency response.

Box Tone is a high-precision filter inspired by the mid- and high-frequency contours of modern and classic converters. As you'd expect, converters are designed to be as flat and balanced as possible, so these effects are intentionally subtle.

Box Tone captures this nuanced behavior, providing a gentle and refined way to shape the top end of your mix while preserving its natural character.

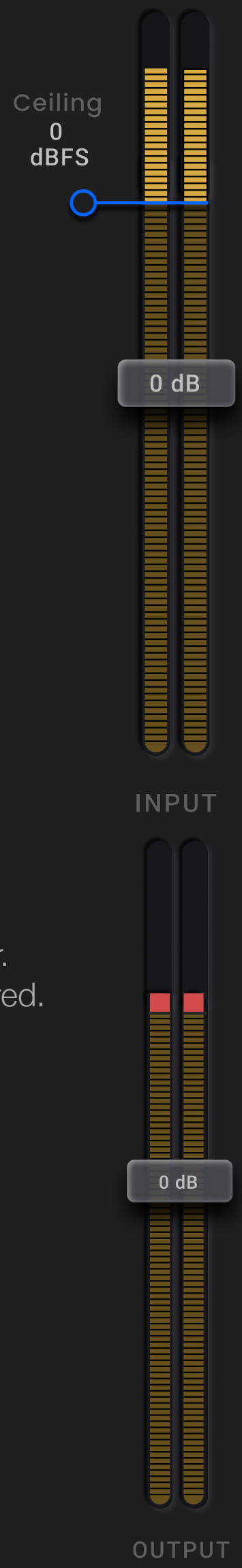
Box Tone in Practice

Use Modern when you want to slightly contour the signal around 8kHz and above and slightly low-pass the signal. Use Classic when you want to slightly low pass the signal but maintain high frequencies. Use Flat when you don't want any high frequency contour or ultra-sonic filtering.

07 | INPUT & OUTPUT TRIM

Input Trim

Input trim is applied to the input signal. The input meter shows the input signal + input trim. The gain knob floats over the input meter and allows for +/- 12dB of gain.



Output Trim

Output trim is applied to the entire plugin output after the mixer.
If the output exceeds 0.0dBFS then the output meter will turn red.

Trim Link

Trim Link allows you to maintain perceived loudness when changing input or output trim. When turning the input trim up by 1dB, Gain Link will turn the output down by 1dB. This function is helpful when working on low-level sources that need additional gain to get to the clip point. Gain Link is helpful when Gold Clip is instantiated on an individual track in a mix or when you are trying to maintain perceived loudness but push the mix up to the clip point.

Input/Output Trim Link in Practice

Turn Trim Link and Gold/Unity ON. Use the Clip Ceiling and/or Trims to lightly clip the signal, then apply compression with Gold and Alchemy. This setup lowers the peaks of the track without changing the perceived level, thus creating more headroom in the master. I generally use linked I/O trims on the mix bus and the ceiling control on individual tracks. If you want to offset the trim links, you can drag the faders while holding down the Shift key. Shift + Option click to reset the offset.

Trim Link Offset

SHIFT and drag your mouse to change the relationship of the linked faders.
SHIFT + OPTION + Click to reset the offset value to 0dB.

Mixer (GOLD CLIP ONLY)

With almost all DAWs running at 32bit float you have 1528dB of dynamic range to work with. Your DAW has all the bandwidth it needs to handle parallel processing, but it's your responsibility to manage your session gain-staging and not overload your mix bus or plugins. You should try to keep Gold Clip's output below 0.0dBFS, but it can go well above that if you choose. The dry signal is not oversampled - it is a copy of the plugin input before oversampling.

Dry / Wet Link (GOLD CLIP ONLY)



When Link is on, the faders operate like a traditional dry/wet fader found on most plugins. When Link is off, the dry and wet faders work independently, making Gold Clip a true parallel processor. When unlinked, you can make the signal twice as loud by setting both the dry and wet fader to 0.0 dBFS. This will most likely clip devices after Gold Clip. When unlinked, you must watch the output meter to make sure you are not creating signals louder than 0.0 dBFS.

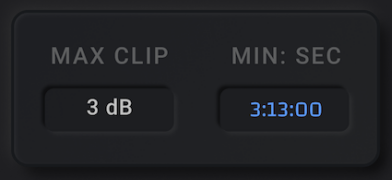
Delta (GOLD CLIP ONLY)



Delta phase flips the input and oversampled output signal, showing you the difference. This function can help you hear the distortion the clipper is adding, the low-level gain that Gold is adding, or the peak information Alchemy is reducing. However, this function is misleading because you are hearing both what is added (Gold and clipping) and what is taken away (Box Tone and Alchemy). In addition, when in minimum phase oversampling mode, you will hear the phase shift from the minimum phase filters. This will make you think you are changing that part of the signal, but you are simply hearing the phase shift of minimum phase filtering.

Mixer in Practice

1) Gold Clip has a true parallel mixer. You can keep the dry signal at 100% and blend in clipping at lower levels.
2) Turn Delta on, and Gold, Alchemy, and Box Tone off. Then use linear phase oversampling and listen to the 100% wet signal while setting clipper. This will help you hear how the clipping is affecting transients.
3) Use clipping, Gold and Alchemy as you desire. Unlink dry and wet and blend in dry transient information from the bottom-up.



Clip Clock (GOLD CLIP ONLY)

Clip Clock is a unique feature in Gold Clip, designed to give you precise control over your clipping and Gold processing. It offers two essential functions:

Max Clipping Amount

Clip Clock monitors the maximum amount of clipping in your track. If you clip by 5 dB and that's more than you intended, the MAX CLIP field will clearly display it, allowing you to make adjustments as needed.

Max Clipping Location

Clip Clock records the exact MIN:SEC where the maximum clipping occurred, making it easy to pinpoint the location. Even if the plugin window is closed, Clip Clock continues to monitor your clipping. To evaluate where your clipper is working the hardest, navigate to the MIN:SEC in your song, take a listen, and decide if the clipping is too much or just right.

You can reset MAX CLIP or MIN:SEC anytime by double-clicking on them.

This tool is simple yet invaluable for refining your clipping and Gold processing. Remember: less clipping leads to cleaner Gold processing, as the two are inversely related. That's why Clip Clock exists—to help you fine-tune your clipping and maximize the impact of Gold.

Clip Clock in Practice

Loop the section of the song that clip clock shows having the maximum clipping. Set the clipper so that the clipping is inaudible or as clean as possible during that section of the song. I have also used Clip Clock timing information to look at other dynamic processors in the same signal chain.

Metering

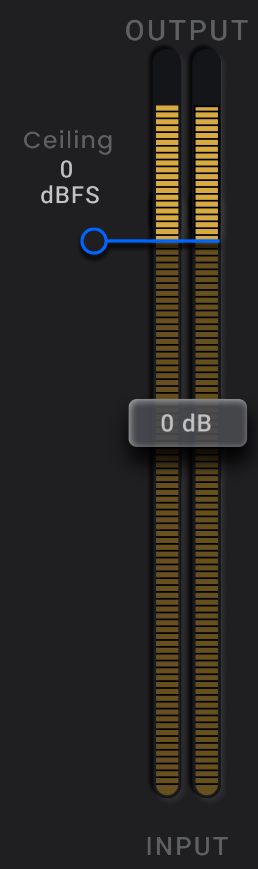
Input Meter

The input meter shows the input signal, as well as the input trim applied to the input signal. The number at the top of the meter displays the maximum amount of clipping. The number at the bottom of the meter provides a continuous readout of Short Term LUFS. When the clipper is turned off, it is important to ensure that the input signal does not exceed 0.0dBFS. If the clipper is off and the input signal rises above 0.0dBFS, the meter will turn red, indicating that you should decrease the input trim.



Output Meter

The output meter shows the sum of the dry and wet signal. The output trim affects the sum of the dry and wet signal. If the output level is above 0.0 dBFS, the meter will illuminate red.



Tool Tips (GOLD CLIP ONLY)



Tool tips can be turned on and off with the "?" button. Hovering over any parameter will bring up a text box describing its function and the quick keys associated with that parameter.

Waveform Display (GOLD CLIP ONLY)

The waveform display window shows Clipping, Plugin Output and Gold Processing.



Clipping – The grey outlined signal above the clip point consists of the input signal and input trim and shows you the peaks that were removed by the clipper.

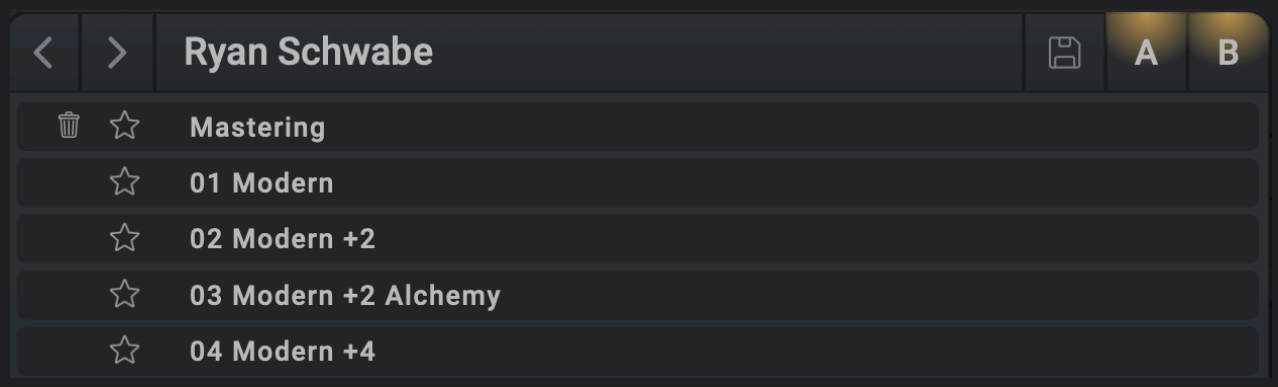
Plugin Output – The black waveform with Gold border shows the plugin output.

Gold Processing – The Gold fill that rises around black waveform shows the nonlinear loudness saturation that is added by Gold processing.

Clipping Window Magnification – Double clicking the waveform scale on the right or left side of the waveform, will change the waveform display scaling from 6dB, 4dB and 2dB.

Presets

Easily create and set your default mixing and mastering presets.



- Click the disk button to save a new preset
- Name the preset and hit enter
- Click the trashcan to delete presets
- Click the Gold Star to set the default preset.*

*Pro Tools has a proprietary default preset system that overrides the "star" functionality in Gold Clip and Gold Clip Track. If you are a Pro Tools user we recommend using their proprietary default preset system to avoid settings conflict.

AB

AB made simple. When you first open Gold Clip, both A and B states are active; they are both lit up. As you make changes to the settings, they are applied to both A and B. Only when you click A or B will they unlink so that you can make more detailed changes to one plug-in state. Option-click A to copy to B or Option-click B to copy to A. You can toggle between A and B by simply clicking A or B repeatedly.

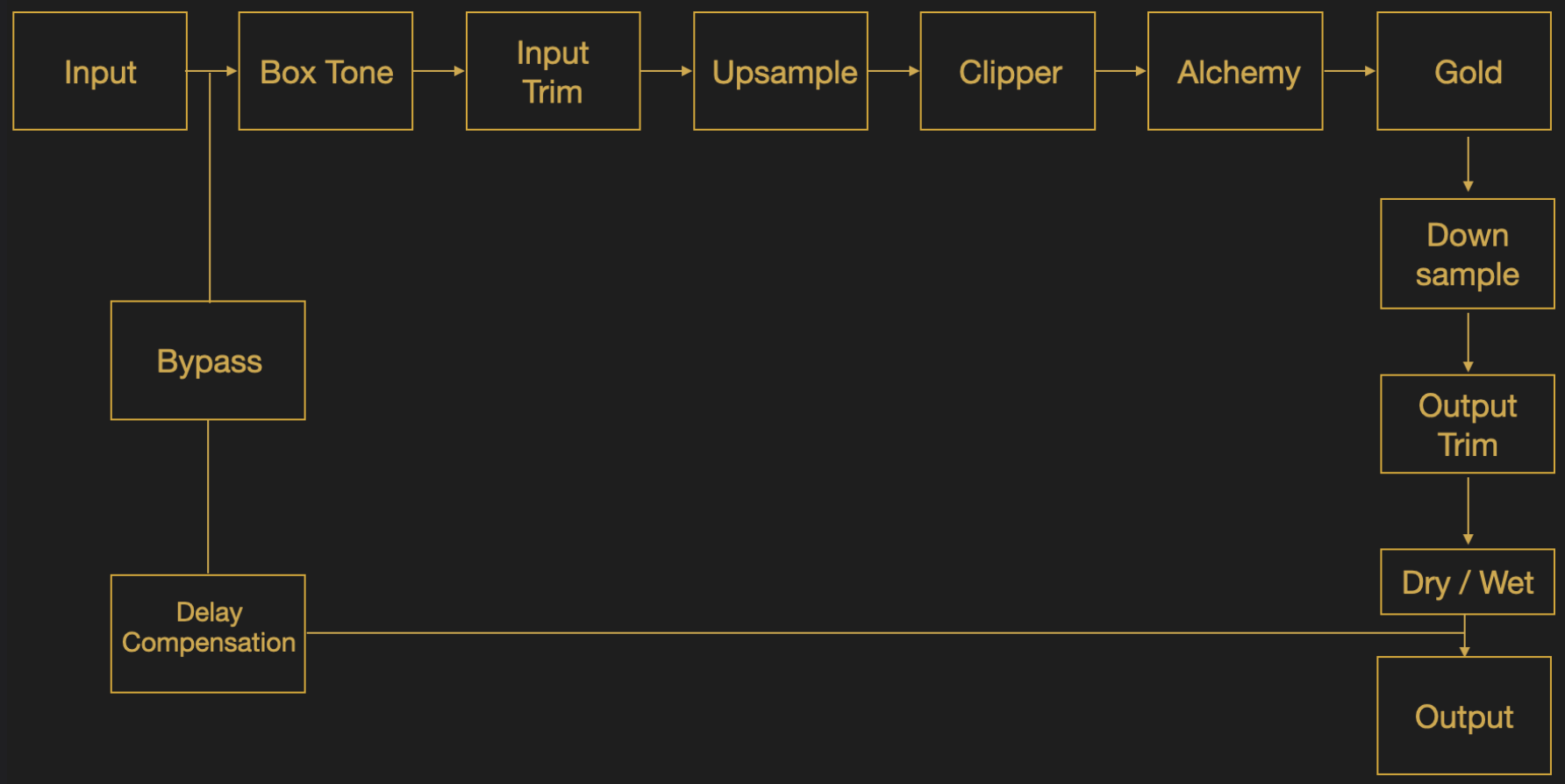
AB in Practice

Alter Gold Clip to the settings that work best for the track and then click B to unlink the A and B settings. Now try different clipper types and amounts of Gold & Alchemy processing. You can click A or B repeatedly to alternate plugin settings. Option-Click A or B to copy the setting to the other plugin state.

Bypass

This is a true bypass that is internally time-aligned with the plugin delay. When automating the bypass, you should use the true bypass rather than the DAW bypass function.

09 | SIGNAL FLOW

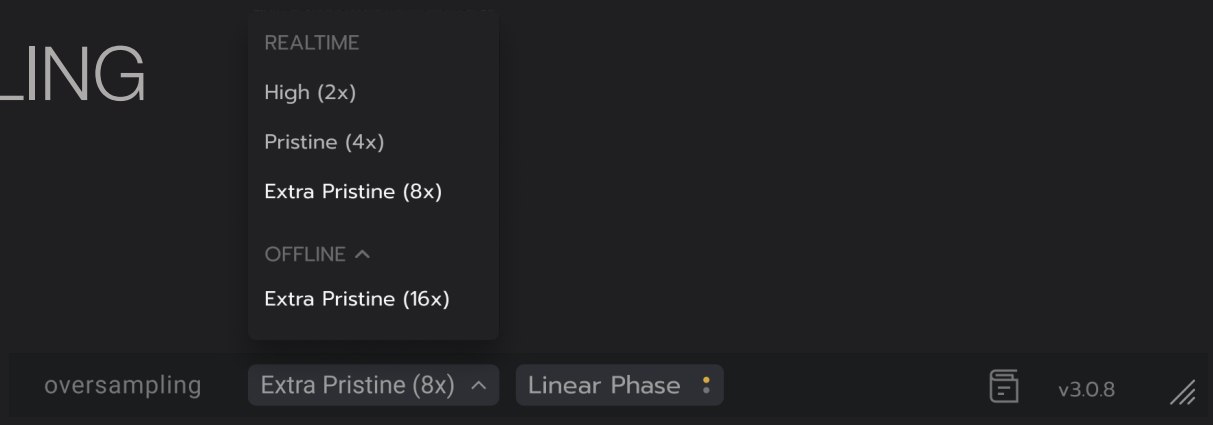


10 | KEY COMMANDS



1 - Box Tone - Command click to bypass
2 - Clipper - Command click to bypass
3 - Gold - Command click to bypass
4 - Clip Clock - Double-click numeric values to reset
5 - Gold Amount - Command click to bypass
6 - Alchemy - Command click to bypass
7 - Output Trim - Option-click to return to 0
Shift-Click offset linked trims
Shift-Option-Click to reset offset to 0
8 - Mixer - Option-click fader to return to 100% wet
9 - Output Peak Readout - Double-click to reset
10 - Input Trim - Option-click to return to 0
Shift-Click offset linked trims
Shift-Option-Click to reset offset to 0
11 - Clipping Numeric - Double-click to reset
12 - Schwabe Digital Logo - Click for plugin options

10 | OVERSAMPLING



Oversampling Rate

High - 4x (44.1kHz & 48kHz), 2x (88.2kHz and 96kHz), 1x (192kHz+)

Pristine - 8x (44.1kHz & 48kHz), 4x (88.2kHz and 96kHz), 2x (192kHz+)

Extra Pristine - 16x (44.1kHz & 48kHz), 8x (88.2kHz and 96kHz), 4x (192kHz+)

Gold Clip allows you to select separate sample rates for real time processing and offline bouncing.

Oversampling Type

Linear Phase Minimum Phase

Linear Phase – Uses a custom linear phase oversampling technique. This option is suggestion if you are doing parallel processing within Gold Clip.

Minimum Phase – Uses a custom minimum phase over sampling technique. This option is sometimes better for low frequency transient information that does not require parallel processing. When in minimum phase mode the delta function will present some of the phase rotation that is common with minimum phase oversampling. Phase rotation is completely normal behavior and is common in many audio processors.

Anti-Overshoot Technology

Our anti-overshoot technology ensures your audio remains below the ceiling at all times, preventing any peaks from shooting above after downsampling.

Oversampling is a dense and often over-debated topic. We chose to make our oversampling as robust as possible, while still keeping it simple. Nerd out and try all the options or just leave it at its default and let it do its thing.

12 | COMPATIBILITY

We highly encourage you to use modern computers with Gold Clip. Computers older than 2018 are most likely not supported. Please test Gold Clip on your system before purchasing.

MAC:

Mac OS 10.13 or higher

Native Silicon Support

Apple Intel - Intel Core 2 Duo or higher, 4GB RAM minimum

PC:

Windows 11 (recommended)

Windows 10

Intel Core 2 Duo / AMD Athlon 64 X2 or higher, 4GB RAM minimum

AUDIO SESSION COMPATIBILITY:

16bit, 24bit or 32bit

44.1kHz, 48kHz, 88.2kHz or 96kHz sessions (192kHz not supported)

VST3, AU AAX